Manual 7: Installing the PC

For whoever prepares the registration laptop

Python, the apps, the email account, the Wi-Fi and the USB stick: everything the laptop needs before Manual 1 can start. Do it a week early, never on race morning.

What you need

item	why
A Windows 10 or 11 laptop (macOS and Linux also work)	runs the race desk, the easy desk and the check-in server
Its charger, and an extension lead	the server runs all day
A USB stick on a lanyard	the backup copy of the check-in record, written after every scan
The marquee Wi-Fi router	the phones talk to the laptop over it; it does not need the internet
Internet access, the week before	to install Python, send the letters and read the replies
The apps folder, *trail_race_apps*	from the zip

1. Install Python

1. Go to *python.org*, Downloads, and download the latest Python 3 for Windows (3.11 or newer).
2. Run the installer. On the first screen, tick **Add python.exe to PATH**. This is the step people miss.
3. Press **Install Now**. Tkinter, which draws the race desk's window, comes with it.
4. Open **Command Prompt** (Start, type *cmd*) and check:

```
python --version
```

It should print Python 3.11 or newer. If it says Python was not found, run the installer again and tick **Add python.exe to PATH**.

2. Put the apps in place

1. Unzip *trail_race_apps.zip* to a short path with no spaces, for example *C:\trail_race_apps*.

2. Inside you should see the folders *input*, *output*, *racedesk* and *static*, the *.py* programs, *config.ini* and the Word letters.

folder	what lives there
input	the EntryNinja export and the slimmed working file. Never emptied.
output	everything the apps make. Clean system empties it.
racedesk	the two windows: race_desk.py and easy.py, and their settings
static	the phone apps, served by the check-in server
manuals, novel	these manuals and the novel, with the tests that photograph them

3. Install the libraries

In Command Prompt:

```
cd C:\trail_race_apps
python -m pip install reportlab openpyxl cryptography "qrcode[pil]"
```

That is everything the race needs. Only to rebuild the manuals or the novel from their screen tests, also:

```
python -m pip install playwright
python -m playwright install chromium
```

4. A first run

Still in Command Prompt, run the smoke test once. It empties the output folder, then makes a label sheet, a calibration page, an invitation and a picture card from the sample export:

```
python reset_and_test.py --yes --test
```

> ***Never run that once real collection has started: emptying output removes the check-in record.***

Then start the race desk:

```
python racedesk\race_desk.py
```

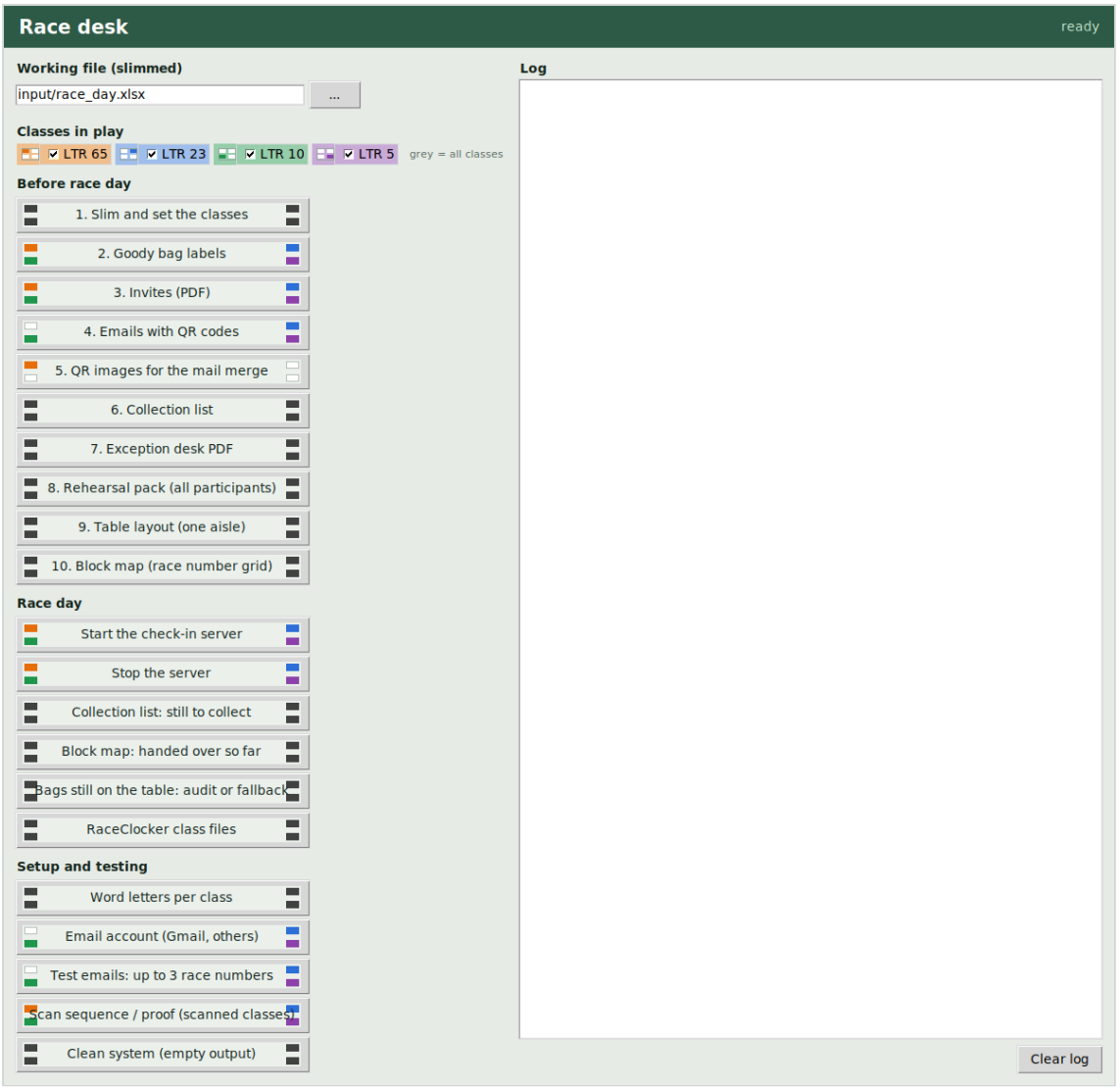


Figure 1. *The race desk, ready. If this window appears, Python, Tkinter and the apps are installed.*

You can also double-click *race_desk.py* and *easy.py* in the *racedesk* folder. Make desktop shortcuts to both.

5. The email account

The letters go from a race address, not a personal one.

1. In Google, sign in to the race address. Security, 2-Step Verification: switch it on.
2. Security, **App passwords**: make one called *race desk*. Google shows 16 letters.
3. In Gmail, Settings, See all settings, Forwarding and POP/IMAP: **Enable IMAP** (the reply reader needs it).
4. On the race desk: **Email account (Gmail, others)**, press **Gmail / Workspace**, fill in the address and the 16 letters, **Send test** to yourself.

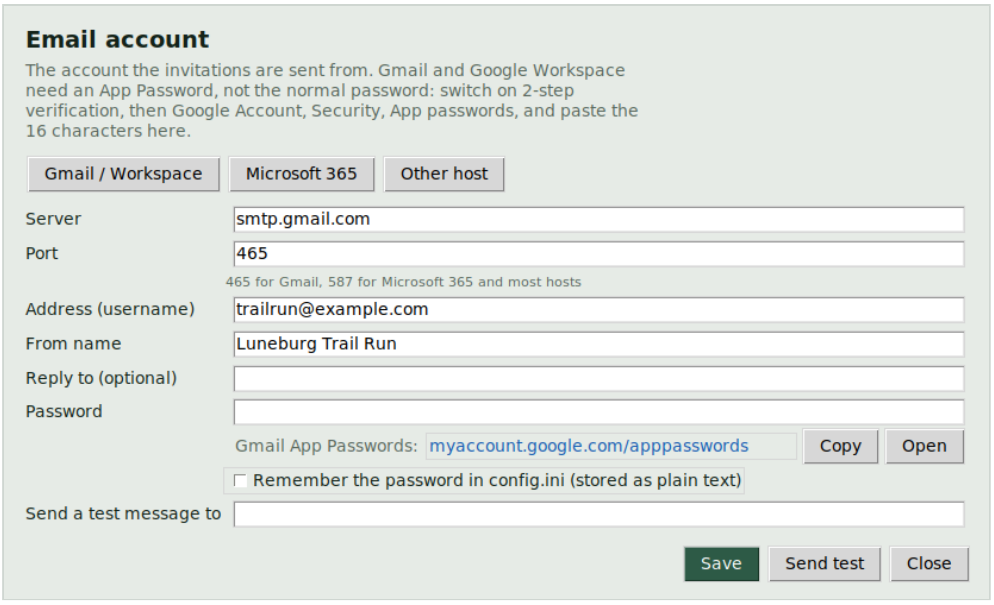


Figure 2. *The account screen. Leave* ***Remember the password*** *unticked on a shared laptop: the desk asks once per session instead.*

6. config.ini

Open *config.ini* in Notepad. The parts you may need to change:

section	what
[csv]	which EntryNinja columns the apps read. id_column = Member # unless check_ids.py says otherwise
[smtp]	the email account (the race desk's Email account screen writes this for you)
[send]	the subject line; paid_only
[collection]	the reply deadline, collection times, where the exception desk is, the goody bag note, the sign-off
[imap]	where replies are read from; blank means worked out from [smtp]

7. The marquee: Wi-Fi, firewall and phones

1. Connect the laptop to the marquee router. Give it a fixed address in the router if you can, so the phone codes never change.
2. Start the check-in server once (easy desk, Import and start; or race desk, Start the check-in server).
3. Windows asks whether Python may use the network. Choose **Private networks** and **Allow access**. If you missed it: Windows Security, Firewall, Allow an app, Python, tick Private.
4. The server prints where the phones should go:

```
Open on volunteer phones:
  https://192.0.2.2:8443/
  https://lueneburg-trail-run.local:8443/
PIN: none - the phones only ask for a name
Exception desk (authorisation tokens): https://192.0.2.2:8443/exception   [only on this PC unless --
desk-pin is set]
Bags still on the table (audit / paper fallback): https://192.0.2.2:8443/leftover
```

5. It also writes two codes into the output folder for the phones to scan. Print them.



Figure 3. *The scanner app.*



Figure 4. *The table-walk app.*

6. On a phone, scan the first code. The browser warns that the connection is not private: the laptop made its own certificate. Choose **Advanced**, then **Proceed**. Each phone asks once.
7. Allow the camera when the scanner asks.

The laptop also answers to the name *lueneburg-trail-run.local* on iPhones and other PCs. Most Android phones cannot look up that kind of name; the codes carry the number, which every phone can open.

8. The USB stick

1. Plug it in and note its drive letter, for example *E:*.
2. On the race desk's check-in server screen (or the race desk settings the easy desk reads), set **Backup drive or folder** to it.
3. Tie it to the table leg. If it comes out, alert 08 appears on the laptop and a red banner on every phone; plug it back in.

9. The week before: the checklist

check	how
Python and the apps	the race desk opens
The export slims	Manual 1, step 1, with this week's export
The email account	Send test reaches you
The letters	three test letters reach you and read well on a phone
The phones	a volunteer's phone opens the scanner on the marquee Wi-Fi, through the certificate warning
The rehearsal	the rehearsal pack is printed, and scanned against the rehearsal server (Manual 3)
The backup	the USB stick is set, and alert 08 clears when it goes back in
The paper	Manual 6, part A
Clean before the day	after all the practice: Clean system, with "also clear the check-in record"

When something goes wrong

what you see	what to do
python is not recognised	reinstall Python with **Add python.exe to PATH** ticked
No module named reportlab (or openpyxl, qrcode)	run the pip line in section 3 again
The race desk window does not open on Linux	install tkinter: *sudo apt install python3-tk*
Phones cannot open the scanner	same Wi-Fi? Firewall allowed on private networks? Laptop awake?
The certificate warning every time	the laptop's address changed: delete *cert.pem* and *key.pem* in output, restart the server, and fix the address in the router
Gmail refuses the password	use an App Password, not the account password